

Differential Amplifier Design

Specifications

Technology : 4PDK180 (180nm)

$$V_{DD} = 1.8V$$

$$A_v \geq 100V/V \text{ (40dB)}$$

$$GB \geq 5MHz$$

$$C_L = 10pF$$

$$\text{Slew rate} \geq 5V/\mu\text{sec}$$

$$I_{CMR} = 0.8V - 1.6V$$

$$\text{Power dissipation} \leq 0.3mW$$

Tail current (I_{SS})

We need tail current when M_1 is off and M_2 is ON so that we can achieve desired slew rate (entire I_{SS} flows through M_2 to charge C_L)

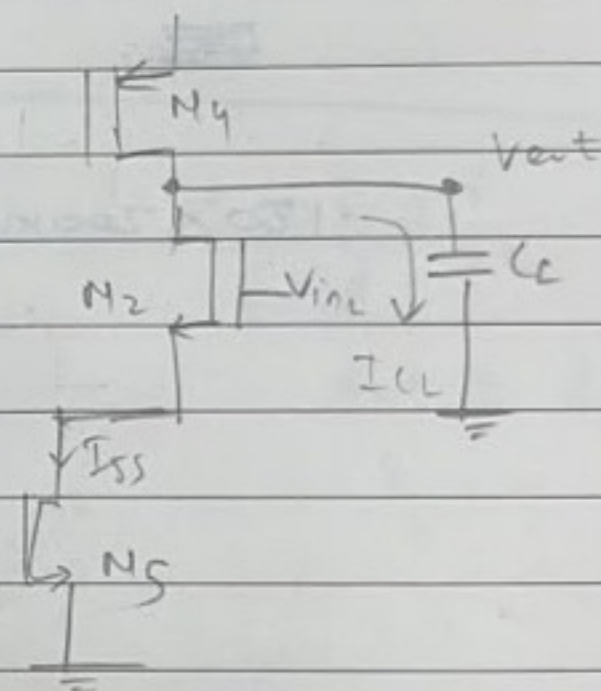
$$\therefore I_{CL} = C \frac{dV_{out}}{dt} = I_{SS}$$

$$\text{also } \frac{dV_{out}}{dt} = \text{slew rate}$$

$$I_{SS} = S.R \times C_L$$

$$= 5 \times 10^6 \times 10 \times 10^{-12}$$

$$I_{SS} = 50\mu A$$



$\therefore$ Now we know DC current through all MOSFETs

$$I_{M1}, I_{M3} = 25\mu A$$

$$I_{M2}, I_{M4} = 25\mu A$$

$$I_{M5}, I_{M8} = 50\mu A$$

For power dissipation

$$P \leq 0.3 \text{ mW}$$

$$P = V_{OD} I_{SS}$$

$$\therefore 1.8 \times I_{SS} \leq 0.3 \text{ mW}$$

$$I_{SS} \leq 166.67 \mu\text{A}$$

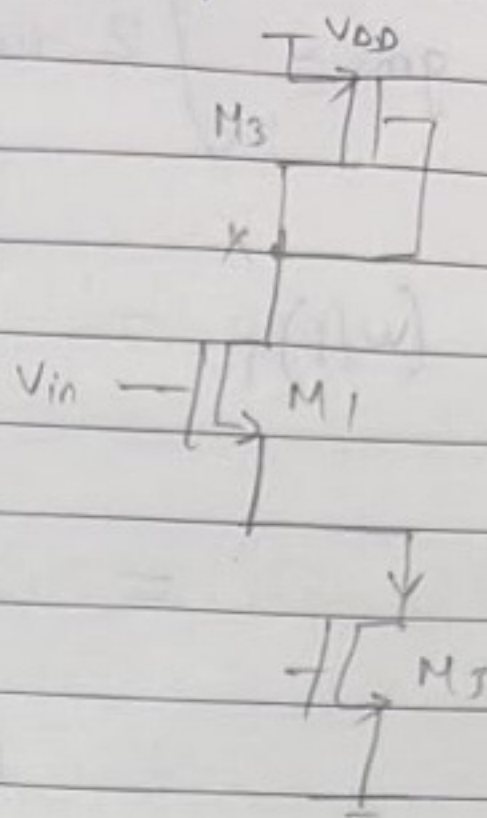
$$\therefore I_{SS} = 50 \mu\text{A} \quad \text{chosen}$$

2.7

Calculation for $(w/L)_3$ and $(w/L)_4$

∴ we need to set upper limit (V_{CMR+}) for the input to keep M_1 in saturation

if V_{in} increases above that then it will enter triode region



$$V_x > V_{in} - V_{th}$$

$$V_{DD} - V_{S43} > V_{CMR+} - V_{thn}$$

$$1.8 - V_{S43} > 1.6 - 0.48$$

$$V_{S43} \leq 0.68V$$

$$I_{M3} = \frac{1}{2} \mu_{plox} \left(\frac{w}{L} \right)_3 (V_{S43} - V_{thp})^2$$

$$25 \mu = \frac{1}{2} 60 \mu \left(\frac{w}{L} \right)_3 (0.68 - 0.45)^2$$

$$\left(\frac{w}{L} \right)_3 = 37$$

$$V_{S43} = V_{S44}, \quad (w/L)_3 = (w/L)_4 = 37$$

Calculation for $(w/L)_1$ and $(w/L)_2$

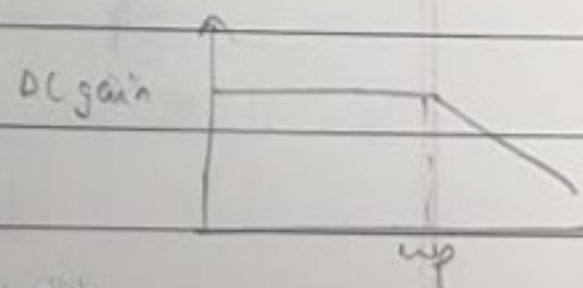
$$DC \text{ gain} = g_{m1} (r_{o2} || r_{o4})$$

$$Pole(\omega) = \frac{1}{RC} = \frac{1}{(r_{o2} || r_{o4}) C_L}$$

$$Pole(f) = \frac{1}{2\pi RC} = \frac{1}{2\pi (r_{o2} || r_{o4}) C_L}$$

$$GB = \frac{g_{m1}}{2\pi C_L}$$

$$g_{m1} = GB \times C_L \times 2\pi, \quad \boxed{g_{m1} = 314.16 \mu S}$$



$$g_{m1} = \sqrt{2 \mu_n C_{ox} \left(\frac{W}{L}\right)_1 I_{M1}}$$

$$\left(\frac{W}{L}\right)_1 = \frac{g_{m1}^2}{2 \mu_n C_{ox} I_{M1}}$$

$$= \frac{9.869 \times 10^{-8}}{2 \times 200 \times 10^{-6} \times 25 \times 10^{-6}}$$

$$= 0.57 \approx 7$$

$$\therefore \left(\frac{W}{L}\right)_1 = 7$$

M1 and M2 are identical $\therefore \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 7$

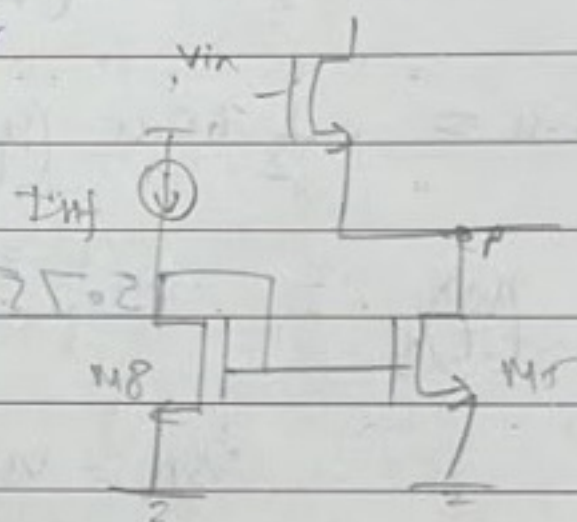
4.7 Calculation for $\left(\frac{W}{L}\right)_5$ and $\left(\frac{W}{L}\right)_8$

We need to set lower

limit for V_{in} (P(NR-))

to ensure that M7

stays in saturation $I_1 = 25 \mu A$



$$\therefore V_{DS5} \geq V_{GS5} - V_{th}$$

$$V_{in} - V_{GS1} \geq V_{GS5} - V_{th}$$

$$0.8 - \left(0.48 + \sqrt{\frac{2I_1}{\mu_n C_{ox} \frac{W}{L}}}\right) = V_{GS5} - 0.48$$

$$V_{GS5} = 0.64569V$$

$$\left(\frac{W}{L}\right)_5 = \frac{I_{M5} \Delta 2}{\mu_n C_{ox} (V_{GS} - V_{th})^2} = 12.1418 \approx 13$$

$$\left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_8 = 13$$